Voice-Based Depression Detection

Capstone Review Summary — Plain-Language Explainer

1. What This Project Does

The system listens to a recording of a clinical interview and tries to guess how depressed the speaker is, using only their voice — pitch, pauses, tone, and energy — without reading or understanding what they actually said. It gives two answers: a category (Minimal/Mild, Moderate, or Moderately Severe/Severe depression) and a score from 0 to 24 on the PHQ-8 scale, the standard clinical depression questionnaire.

The data comes from DAIC-WOZ, a well-known research dataset of 183 real clinical interviews, each already labeled with a real PHQ-8 score by clinicians.

2. The Three Models We Compared — How Each One Actually Works

Model 1 — PCA + Gradient Boosted Trees (our best-performing model)

Step 1 — Turning voice into numbers. Every interview gets converted into about 922 separate measurements: 88 standard acoustic descriptors (pitch, voice jitter, loudness, breathiness, and similar), 66 numbers summarizing pause and rhythm patterns, and 768 numbers from a pretrained speech AI (Wav2Vec2) that has already listened to huge amounts of human speech and learned subtle patterns no one bothered to name. Each interview becomes one long list of 922 numbers.

Step 2 — Compressing that list (PCA). We only have about 130 interviews to train on, but each one has 922 numbers — far more measurements than examples. That's a problem: a model given more knobs to tune than people to learn from tends to just memorize each individual's personal voice quirks instead of learning what depression actually sounds like. PCA (Principal Component Analysis) solves this by finding the 25 combinations of those 922 numbers that capture the most meaningful variation — a bit like summarizing a long book down to its 25 most important plot points, without losing the story.

Step 3 — Gradient Boosted Trees. This is a committee of roughly 100 very simple decision trees, built one after another. The first tree makes a rough, imperfect guess for every interview. The second tree is built specifically to fix whatever the first one got wrong. The third fixes what's still wrong after the first two, and so on — each tree specializing in correcting the mistakes of the ones before it. By the end, their combined vote is far more accurate than any single tree alone. Because each tree is kept deliberately shallow (only 3 questions deep), the whole committee stays simple enough not to memorize individual voices, which is exactly what we need with this little data.

Final step. The trees' combined vote produces a probability for each of the 3 severity categories, and the highest one is the model's answer.

Model 2 — DualBranchDepressionNet (the deep learning model)

Two separate streams of information. This model keeps two kinds of voice information separate at first, instead of blending everything into one list like Model 1 does. "Branch A" carries the traditional acoustic measurements (pitch, pauses, loudness). "Branch B" carries the 768-number deep AI representation of speech. Each branch is passed through its own small neural layer that reshapes and normalizes it.

A learned "dimmer switch" between the two streams (gating). Rather than always trusting both streams equally, the network has a small learned mechanism that acts like a dimmer switch for each one — for every moment in the interview, it decides how much weight to give the traditional acoustic signal versus the deep AI signal, and blends them accordingly. This lets the model lean on whichever signal is more informative at that specific moment.

Reading the interview in both directions (BiLSTM). The blended signal, frame by frame across the whole interview, is fed into a type of neural network called a Bidirectional LSTM. "Bidirectional" means it processes the interview both forward and backward in time, building up a running memory of what's happened so far from both directions — similar to how re-reading a sentence backward can sometimes reveal context you missed reading it only forward. This is what lets the model reason about patterns that unfold over time, like a gradual flattening of tone, rather than only single-moment snapshots.

Deciding which moments mattered most (self-attention). Instead of only looking at the very last moment of the interview for its final answer, the model learns to look back across every moment and assign each one an importance weight — some moments (a long uncomfortable pause, a flat, toneless answer) get weighted more heavily than others (routine small talk). The final "summary" of the whole interview is a weighted average built from these importance scores, not just whatever happened last.

Three separate answers from one summary. That single summary is then fed into three small, separate output layers, each trained for a different specific question: which of the 3 severity categories fits best, what the continuous 0–24 PHQ-8 score should be, and a direct yes/no on whether the person is depressed at all. Training all three at once helps the model build a richer, more useful internal summary than training on just one question alone.

One more step — confidence calibration. Neural networks are notoriously overconfident — they'll often say "99% sure" even when they're wrong. After training is completely finished, a separate small step (temperature scaling) adjusts how loudly the model states its confidence, without ever changing its actual answer, so the confidence numbers you see are more honest.

Model 3 — Hierarchical Two-Stage Classifier

The core idea: ask an easier question first. Sorting someone into one of 3 categories in a single step is hard when one of those categories (Severe) has only a handful of real examples to learn from. This model instead breaks the decision into two much easier yes/no questions, asked one after another — similar to how a doctor might do a quick general triage before running a specialized test only on patients who screened positive.

Stage 1 — "Is this person fine, or not?" A simple, well-understood model (logistic regression) is trained on just this one binary question, using the same 25 compressed measurements from Model 1's PCA step. Because "fine" (154 people) versus "not fine" (29 people) is a far more balanced, learnable split than trying to separate 3 unevenly-sized groups at once, this stage tends to be fairly reliable.

Stage 2 — "If not fine, is it moderate or severe?" A second model (a Support Vector Machine, which finds the best boundary line separating two groups in the compressed feature space) is trained only on the people Stage 1 flagged as "not fine" — asking it to make just one, more focused decision instead of trying to separate all 3 categories at once.

Combining the two stages. The final probability for each category is built by multiplying the two stages' confidence together — for example, the chance someone is "Severe" is Stage 1's confidence they're "not fine" multiplied by Stage 2's confidence that, given they're not fine, they're on the severe end rather than the moderate end.

3. The Headline Numbers

Model	Accuracy	Macro-F1	PHQ-8 Error
PCA + Gradient Boosted Trees	81–84%	~30%	3.4–3.7 pts
DualBranchDepressionNet (BiLSTM)	76–81%	~29%	3.6–4.2 pts
Hierarchical Two-Stage	74–76%	~28%	5.1 pts

MAE ("PHQ-8 Error") means: on average, the predicted score is off from the real score by that many points, out of 24.

4. Why Good-Looking Accuracy Doesn't Mean It Works Well

154 of 183 people (84%) are basically fine. A system that always guesses "fine" — without listening to the audio at all — would already score around 84% accuracy by luck alone. Our models lean on this same effect: reliably good at recognizing "fine," still weak at flagging real Moderate/Severe cases, since there are only a handful of real examples of those to learn from.

Key honest point for review: the numbers are real, not a bug — they show strength at ruling depression out more than strength at flagging it in. That distinction is the actual finding, not a flaw to hide.

5. Why This Happens — Three Reasons

1. **Too few severe examples.** Only 9 people total are severe — not enough to teach any model that pattern reliably.
2. **Voice alone can't hear everything.** Someone can describe severe distress calmly. Audio-only research typically tops out around 60–78% accuracy for this reason.
3. **Our accuracy is trustworthy.** Every person's audio stays entirely in either training or testing, never both — so we're always tested on voices never heard before. Many published 90%+ results let the same speaker appear on both sides, letting the model cheat.

6. What We'd Do With More Time

Add transcripts, not just audio — combining what was said with how it was said is where the biggest future gains are. **Use a larger modern speech AI model. Look at response-pause timing** as its own signal. **Reframe rare cases as "unusual," not a category to classify** — train only on what "fine" sounds like and flag anything meaningfully different, sidestepping the need for many rare examples.

7. One-Paragraph Summary to Say Out Loud

"Our pipeline is fully leak-free and benchmarked across five different modeling approaches, from deep neural networks to classical machine learning. Because only 9 of our 183 interviews are severe cases, no approach — regardless of complexity — could reliably detect that class, which matches known limitations documented in published DAIC-WOZ research. Our best model reliably rules out depression and estimates PHQ-8 score within about 3.4 points on average, both in line with published audio-only baselines. Reliably detecting moderate-to-severe cases needs more real examples of those cases than this dataset provides — a clear, evidence-based direction for future work."